

Solutions to Problem Set 11

1. (*) Convergence in probability [4 Points]

Let $(\Omega, \mathcal{F}, \mathbf{P})$ a probability space. Suppose that for every $n \in \mathbb{N}$, we have random variables $X_n, Y_n : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, and let $X, Y : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be random variables.

- (a) Suppose that $X_n \xrightarrow[n \rightarrow \infty]{\mathbf{P}} X, Y_n \xrightarrow[n \rightarrow \infty]{\mathbf{P}} Y$. Show that

$$X_n + Y_n \xrightarrow[n \rightarrow \infty]{\mathbf{P}} X + Y.$$

- (b) Let $(a_n)_{n \in \mathbb{N}}$ be a sequence of real numbers with $\lim_{n \rightarrow \infty} a_n = a$. Suppose that $X_n \xrightarrow[n \rightarrow \infty]{\mathbf{P}} x$, for some $x \in \mathbb{R}$. Show that

$$a_n \cdot X_n \xrightarrow[n \rightarrow \infty]{\mathbf{P}} a \cdot x.$$

- (a) Let $\varepsilon > 0$. By the triangle inequality,

$$\text{If } |X_n - X| \leq \frac{\varepsilon}{2}, \quad |Y_n - Y| \leq \frac{\varepsilon}{2} \quad \text{then } |X_n + Y_n - (X + Y)| \leq \varepsilon.$$

In other words:

$$\{|X_n - X| \leq \frac{\varepsilon}{2}\} \cap \{|Y_n - Y| \leq \frac{\varepsilon}{2}\} \subseteq \{|X_n + Y_n - (X + Y)| \leq \varepsilon\}.$$

By taking complements, we see that

$$\{|X_n + Y_n - (X + Y)| > \varepsilon\} \subseteq \{|X_n - X| > \frac{\varepsilon}{2}\} \cup \{|Y_n - Y| > \frac{\varepsilon}{2}\}.$$

Therefore

$$\mathbf{P}[|X_n + Y_n - (X + Y)| > \varepsilon] \leq \mathbf{P}[|X_n - X| > \frac{\varepsilon}{2}] + \mathbf{P}[|Y_n - Y| > \frac{\varepsilon}{2}] \xrightarrow[n \rightarrow \infty]{} 0,$$

where we used in the last display we used $X_n \xrightarrow[n \rightarrow \infty]{\mathbf{P}} X, Y_n \xrightarrow[n \rightarrow \infty]{\mathbf{P}} Y$.

- (b) Since $(a_n)_{n \in \mathbb{N}}$ is convergent, it is bounded, so $|a_n| \leq C$ for every $n \in \mathbb{N}$ for some $C > 0$. Now let $\varepsilon > 0$.

$$\begin{aligned} \mathbf{P}[|a_n X_n - a x| > \varepsilon] &= \mathbf{P}[|a_n X_n - a_n x + a_n x - a x| > \varepsilon] \\ &\leq \mathbf{P}[|a_n| \cdot |X_n - x| > \frac{\varepsilon}{2}] + \mathbf{P}[|x| \cdot |a_n - a| > \frac{\varepsilon}{2}] \\ &\leq \mathbf{P}[|X_n - x| > \frac{\varepsilon}{2C}] + \mathbf{P}[|x| \cdot |a_n - a| > \frac{\varepsilon}{2}] \end{aligned}$$

Both terms converge to zero as $n \rightarrow \infty$, since $X_n \xrightarrow[n \rightarrow \infty]{\mathbf{P}} x$, and $|x| |a_n - a|$ is (deterministically) smaller than $\frac{\varepsilon}{2}$ for all n large enough.

2. Weak law of large numbers

- (a) Let $(X_n)_{n \in \mathbb{N}}$ be i.i.d. real random variables with $X_1 \sim \mathcal{E}(\frac{1}{2})$. Use the weak law of large numbers to show that

$$\frac{1}{n} \sum_{i=1}^n \exp\left(-\frac{X_i}{2}\right) \xrightarrow[n \rightarrow \infty]{\mathbf{P}} \frac{1}{2}.$$

(b) Let $(X_n)_{n \in \mathbb{N}}$ be i.i.d. real random variables such that the law of X_1 has the probability density function

$$f_{X_1}(x) = \begin{cases} \frac{1}{6}, & 0 \leq x < 2, \\ \frac{1}{3}, & 2 \leq x \leq 4, \\ 0, & \text{else} \end{cases}$$

Calculate the value of $a \in \mathbb{R}$ such that

$$\frac{1}{n} \sum_{k=1}^n \sqrt{X_k} \xrightarrow[n \rightarrow \infty]{\mathbf{P}} a.$$

(a) Since $X_1, \dots, X_n$ are i.i.d., so are $\exp(-X_1/2), \dots, \exp(-X_n/2)$. By the weak law of large numbers,

$$\frac{1}{n} \sum_{i=1}^n \exp\left(-\frac{X_i}{2}\right) \xrightarrow[n \rightarrow \infty]{\mathbf{P}} \mathbf{E}\left[\exp\left(-\frac{X_1}{2}\right)\right],$$

provided that

$$\mathbf{E}\left[\exp\left(-\frac{X_1}{2}\right)\right] = \mathbf{E}[\exp(-X_1)] < \infty.$$

The latter is true since $X_1 \geq 0$, so $\exp(-X_1) \leq 1$. We calculate

$$\mathbf{E}\left[\exp\left(-\frac{X_1}{2}\right)\right] = \int_0^\infty \frac{1}{2} e^{-\frac{x}{2}} \cdot e^{-\frac{x}{2}} dx = \frac{1}{2} [-e^{-x}]_0^\infty = \frac{1}{2}.$$

(b) Since $X_1, \dots, X_n$ are i.i.d., so are $\sqrt{X_1}, \dots, \sqrt{X_n}$. By the weak law of large numbers, we have

$$\frac{1}{n} \sum_{i=1}^n \sqrt{X_i} \xrightarrow[n \rightarrow \infty]{\mathbf{P}} \mathbf{E}[\sqrt{X_1}],$$

provided that

$$\mathbf{E}\left[(\sqrt{X_1})^2\right] = \mathbf{E}[X_1] < \infty.$$

The latter is true since X_1 is bounded. We calculate

$$\begin{aligned} \mathbf{E}[\sqrt{X_1}] &= \int_0^2 \frac{1}{6} \sqrt{x} dx + \int_2^4 \frac{1}{3} \sqrt{x} dx = \left[\frac{1}{6} \cdot \frac{2}{3} x^{\frac{3}{2}}\right]_0^2 + \left[\frac{1}{3} \cdot \frac{2}{3} x^{\frac{3}{2}}\right]_2^4 \\ &= \frac{2}{9}(8 - \sqrt{2}). \end{aligned}$$

3. Applications of the Borel-Cantelli lemmas

(a) Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of i.i.d. $\mathcal{E}(\lambda)$ -distributed random variables, with $\lambda > 0$.

i. Show that

$$\limsup_{n \rightarrow \infty} \frac{X_n}{\log(n)} = \frac{1}{\lambda}, \quad \mathbf{P}\text{-a.s.}$$

ii. Show that

$$\liminf_{n \rightarrow \infty} \frac{X_n}{\log(n)} = 0, \quad \mathbf{P}\text{-a.s.}$$

(b) Consider $X_1, X_2, X_3, \dots$ independent random variables such that $\mathbf{P}[X_1 = 0] = 1$, and

$$\mathbf{P}[X_n = n] = \mathbf{P}[X_n = -n] = \frac{1}{2n \log(n)}, \quad \mathbf{P}[X_n = 0] = 1 - \frac{1}{n \log(n)}, \quad n \geq 2.$$

Show that $\frac{1}{n} \sum_{i=1}^n X_i$ converges to 0 in $\mathbf{P}$ -probability, but not $\mathbf{P}$ -a.s..

- (a) First, we recall that the cumulative distribution function of $\mathcal{E}(\lambda)$ fulfills $\mathcal{E}(\lambda)[(x, \infty)] = 1 - F_{\mathcal{E}(\lambda)}(x) = e^{-\lambda x}$ for $x > 0$.

i. Let $\epsilon > 0$. We calculate

$$\mathbf{P} \left[X_n > \frac{\log(n)}{\lambda} (1 \pm \epsilon) \right] = e^{-\lambda \frac{\log(n)}{\lambda} (1 \pm \epsilon)} = \frac{1}{n^{1 \pm \epsilon}}.$$

Now note that the events $X_n > \frac{\log(n)}{\lambda} (1 \pm \epsilon)$ are independent, and $\sum_{n=1}^{\infty} \frac{1}{n^{1 \pm \epsilon}} < \infty$ for every $\epsilon > 0$, whereas $\sum_{n=1}^{\infty} \frac{1}{n^{1 - \epsilon}} = \infty$ for every $\epsilon > 0$. By the first resp. second Borel-Cantelli lemma, we see that $\mathbf{P}[\Omega_{\epsilon}^+] = 1$, and $\mathbf{P}[\Omega_{\epsilon}^-] = 1$, where

$$\Omega_{\epsilon}^+ = \bigcup_{n=1}^{\infty} \bigcap_{m \geq n} \left\{ X_m \leq \frac{\log(m)}{\lambda} (1 + \epsilon) \right\}, \text{ and } \Omega_{\epsilon}^- = \bigcap_{n=1}^{\infty} \bigcup_{m \geq n} \left\{ X_m > \frac{\log(m)}{\lambda} (1 - \epsilon) \right\}.$$

Now observe that on the event $\Omega' = \bigcap_{\epsilon \in (0, \infty) \cap \mathbb{Q}} \Omega_{\epsilon}^+ \cap \bigcap_{\epsilon \in (0, \infty) \cap \mathbb{Q}} \Omega_{\epsilon}^-$, we have indeed $\limsup_{n \rightarrow \infty} \frac{X_n}{\log(n)} = \frac{1}{\lambda}$, and the event Ω' has $\mathbf{P}$ -probability 1 (again by a union bound).

- ii. It is clear that $\liminf_{n \rightarrow \infty} \frac{X_n}{\log(n)} \geq 0$, $\mathbf{P}$ -a.s., since ($\mathbf{P}$ -a.s.) we have $X_n > 0$ for all $n \in \mathbb{N}$. Now let $\varepsilon > 0$. Note that

$$\mathbf{P}[X_n < \varepsilon \log(n)] = 1 - e^{-\lambda \varepsilon \log(n)} = 1 - \frac{1}{n^{\lambda \varepsilon}} > c,$$

uniformly in n , and the events $\{X_n < \varepsilon\}$ are independent. By the second Borel-Cantelli lemma, observing that $\sum_{n=1}^{\infty} \mathbf{P}[X_n < \varepsilon \log(n)] = \infty$, we see

$$\mathbf{P} \left[\bigcap_{n=1}^{\infty} \bigcup_{m \geq n} \left\{ \frac{X_m}{\log(m)} < \varepsilon \right\} \right] = 1.$$

Call $\Omega_{\varepsilon} = \bigcap_{n=1}^{\infty} \bigcup_{m \geq n} \left\{ \frac{X_m}{\log(m)} < \varepsilon \right\}$. For every $\omega \in \Omega_{\varepsilon}$ we have

$$\liminf_{n \rightarrow \infty} \frac{X_n(\omega)}{\log(n)} \leq \varepsilon.$$

Again by a union bound, we now see that $\mathbf{P}[\bigcap_{\varepsilon \in (0, \infty) \cap \mathbb{Q}} \Omega_{\varepsilon}] = 1$, and for every $\omega \in \bigcap_{\varepsilon \in (0, \infty) \cap \mathbb{Q}} \Omega_{\varepsilon}$, we have $\liminf_{n \rightarrow \infty} \frac{X_n(\omega)}{\log(n)} = 0$.

- (b) We show that $\frac{1}{n} \sum_{i=1}^n X_i$ converges to 0 in $\mathbf{P}$ -probability: Indeed, fix $\varepsilon > 0$. Note that

$$\text{Var}[X_m] = (+m)^2 \frac{1}{2m \log(m)} + (-m)^2 \frac{1}{2m \log(m)} = \frac{m}{\log(m)}$$

for $m \geq 2$, and $\text{Var}[X_1] = 0$. We have

$$\begin{aligned} \mathbf{P} \left[\left| \frac{1}{n} \sum_{i=1}^n X_i \right| > \varepsilon \right] &\leq \frac{\sum_{i=1}^n \text{Var}[X_i]}{n^2 \varepsilon^2} = \frac{1}{n^2 \varepsilon^2} \sum_{m=2}^n \frac{m}{\log(m)} \\ &\leq \frac{1}{n^2 \varepsilon^2} \frac{1}{\log(n)} \underbrace{\sum_{m=2}^n m}_{= \frac{1}{2} n(n-1)} \rightarrow 0, \end{aligned}$$

as $n \rightarrow \infty$, where we used Chebyshev's inequality (and $\log(n) \geq \log(m)$, so $\frac{1}{\log(m)} \leq \frac{1}{\log(n)}$ for $m \geq n$). We now show that $\frac{1}{n} \sum_{i=1}^n X_i$ does not converge $\mathbf{P}$ -a.s.. To that end, we consider $A_n = \{|X_n| \geq n\}$, which are independent events. We now note that

$$\mathbf{P}[A_n] = \frac{1}{n \log(n)}, \text{ and } \sum_{n=2}^{\infty} \frac{1}{n \log(n)} = \infty,$$

since $\int_2^\infty \frac{1}{x \log(x)} dx = \lim_{x \rightarrow \infty} (\log |\log(x)| - \log |\log(2)|) = \infty$. By the second Borel-Cantelli lemma, we see that

$$\mathbf{P} \left[\bigcap_{n=1}^{\infty} \bigcup_{m \geq n} A_m \right] = 1.$$

For any $\omega \in \bigcap_{n=1}^{\infty} \bigcup_{m \geq n} A_m$, we see that $|X_n(\omega)| \geq n$ for infinitely many n . Suppose now $\frac{1}{n} \sum_{i=1}^n X_i(\omega)$ converges to 0, then for every $\varepsilon > 0$, we have $|\sum_{i=1}^n X_i| \leq n\varepsilon$ for every $n \geq n_0(\omega)$. This however would imply

$$|X_{n+1}| \leq \left| \sum_{i=1}^{n+1} X_i \right| + \left| \sum_{i=1}^n X_i \right| \leq 2(n+1)\varepsilon.$$

Choosing $\varepsilon = \frac{1}{4}$, this is a contradiction to $|X_n(\omega)| \geq n$ for infinitely many n . Since $\bigcap_{n=1}^{\infty} \bigcup_{m \geq n} A_m$ has probability 1, its complement has probability 0, so the set of ω for which $\frac{1}{n} \sum_{i=1}^n X_i(\omega)$ converges to 0 has $\mathbf{P}$ -probability 0.

4. (*) Convergence in distribution

[4 Points]

Consider sequences of real random variables $(X_n)_{n \in \mathbb{N}}$ and $(Y_n)_{n \in \mathbb{N}}$ with probability density functions

$$(i) \ f_{X_n}(x) = \lambda \left(1 - \frac{\lambda x}{n}\right)^{n-1} \mathbb{1}_{[0, \frac{n}{\lambda})}(x), \text{ where } \lambda > 0,$$

$$(ii) \ f_{Y_n}(x) = \frac{n+1}{n} x^{\frac{1}{n}} \mathbb{1}_{[0,1]}(x),$$

Show that X_n and Y_n converge in distribution as $n \rightarrow \infty$, and determine the respective limit distribution.

We calculate the respective cumulative distribution functions:

(i)

$$\begin{aligned} F_{X_n}(x) &= \int_{-\infty}^x f_{X_n}(t) dt \\ &= \begin{cases} 0, & x < 0, \\ \int_0^x \lambda \left(1 - \frac{\lambda t}{n}\right)^{n-1} dt, & 0 \leq x < \frac{n}{\lambda}, \\ 1, & x \geq \frac{n}{\lambda} \end{cases} \\ &= \begin{cases} 0, & x < 0, \\ \left[-\left(1 - \frac{\lambda t}{n}\right)^n \right]_0^x, & 0 \leq x < \frac{n}{\lambda}, \\ 1, & x \geq \frac{n}{\lambda} \end{cases} \\ &= \begin{cases} 0, & x < 0, \\ 1 - \left(1 - \frac{\lambda x}{n}\right)^n, & 0 \leq x < \frac{n}{\lambda}, \\ 1, & x \geq \frac{n}{\lambda}. \end{cases} \end{aligned}$$

We see that

$$F_{X_n}(x) \xrightarrow{n \rightarrow \infty} (1 - e^{-\lambda x}) \mathbb{1}_{[0, \infty)}(x).$$

This is the cumulative distribution function of an $\mathcal{E}(\lambda)$ -distribution. In other words, $X_n \xrightarrow[n \rightarrow \infty]{d} \mathcal{E}(\lambda)$.

(ii)

$$\begin{aligned} F_{Y_n}(x) &= \int_{-\infty}^x f_{Y_n}(t) dt \\ &= \begin{cases} 0, & x < 0, \\ \int_0^x \frac{n+1}{n} t^{\frac{1}{n}} dt, & 0 \leq x < 1, \\ 1, & x \geq 1 \end{cases} \\ &= \begin{cases} 0, & x < 0, \\ \left[t^{\frac{n+1}{n}} \right]_0^x, & 0 \leq x < 1, \\ 1, & x \geq 1 \end{cases} \\ &= \begin{cases} 0, & x < 0, \\ x^{\frac{n+1}{n}}, & 0 \leq x < 1, \\ 1, & x \geq 1. \end{cases} \end{aligned}$$

We see that

$$F_{Y_n}(x) \xrightarrow{n \rightarrow \infty} \begin{cases} 0, & x < 0, \\ x, & 0 \leq x < 1, \\ 1, & x \geq 1. \end{cases}$$

This is the cumulative distribution function of an $\mathcal{U}([0, 1])$ -distribution. In other words $Y_n \xrightarrow[n \rightarrow \infty]{d} \mathcal{U}([0, 1])$.